

Probability and Statistics Refresher

Two-Hour Crash Course

Topics Overview

1. Probability and Set Theory
 2. Operations with Events
 3. Conditional Probability and Independence
 4. Random Variables and Distributions
 5. Discrete vs. Continuous Random Variables
 6. Normal Distribution and Its Applications
 7. Basic Statistical Inference
 8. Point Estimation and Confidence Intervals
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1. Probability & Set Theory

Key Points

- **Sample space:** Ω , the set of all possible outcomes.
- **Event:** A subset of Ω .
- **Axioms:**
 1. $P(A) \geq 0$ for any event A .
 2. $P(\Omega) = 1$.
 3. If $\{A_i\}$ are pairwise disjoint, then

$$P\left(\bigcup_i A_i\right) = \sum_i P(A_i).$$

Quick Exercise

- Roll a fair six-sided die ($\Omega = \{1, 2, 3, 4, 5, 6\}$). Find $P(\text{roll} < 4)$.

2. Operations with Events

Key Points

- **Union** ($A \cup B$): either A or B (or both).
- **Intersection** ($A \cap B$): both A and B .
- **Complement** (A^c): not A .
- **De Morgan's Laws:**
 - $(A \cup B)^c = A^c \cap B^c$
 - $(A \cap B)^c = A^c \cup B^c$
- **Inclusion-Exclusion Principle:**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Quick Exercise

- If $P(A) = 0.4$, $P(B) = 0.3$, and $P(A \cap B) = 0.1$, find $P(A^c)$, $P(A \cup B)$, etc.
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3. Conditional Probability & Independence

Key Points

- **Conditional Probability:**

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0.$$

- **Independence:**
Two events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

- **Chain Rule:**

$$P(A \cap B \cap C) = P(A)P(B | A)P(C | A \cap B).$$

Quick Exercise

- Two fair coin tosses: Show that “First toss=Heads” and “Second toss=Heads” are independent.
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4. Random Variables & Distributions

Key Points

- A **Random Variable (RV)** is a function mapping outcomes in Ω to real values.
 - **Discrete RV**: Takes values in a countable set.
 - **Continuous RV**: Takes values in an interval of real numbers.
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- **PMF** (discrete): $p_X(x) = P(X = x)$.

- **PDF** (continuous): $f_X(x)$, where

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

- **CDF**: $F_X(x) = P(X \leq x)$.
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- **Mean**:

- Discrete: $E[X] = \sum_x x p_X(x)$.

- Continuous: $E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$.

- **Variance**:

- $\text{Var}(X) = E[(X - E[X])^2]$

- Alternatively, $\text{Var}(X) = E[X^2] - (E[X])^2$.

5. Discrete vs. Continuous RV Examples

Discrete

- **Bernoulli, Binomial, Geometric, Poisson**
 - Example: **Binomial**(n, p):

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

Continuous

- **Uniform, Exponential, Normal**
 - Example: **Exponential**(λ):

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

Quick Exercise

- A fair coin is tossed 5 times. Let X be the number of heads. Find $P(X = 2)$ and $E[X]$.
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6. Normal Distribution & Applications

Key Points

- $X \sim N(\mu, \sigma^2)$:

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

- **Standard Normal:** $Z \sim N(0, 1)$. Any $X \sim N(\mu, \sigma^2)$ can be transformed via

$$Z = \frac{X - \mu}{\sigma}.$$

- **68-95-99.7 Rule:** ~68% within $\pm 1\sigma$, 95% within $\pm 2\sigma$, 99.7% within $\pm 3\sigma$.
 - **Central Limit Theorem (CLT):** For large n , the sample mean $\bar{X}$ of i.i.d. RVs is approximately normal.
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Quick Exercise

- If $X \sim N(100, 16)$, find $P(90 \leq X \leq 110)$ using Z-scores or tables.
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7. Basic Statistical Inference

Key Points

- **Population vs. Sample:**
 - Population parameters: μ, σ .
 - Sample statistics: $\bar{X}, s$.

- **Sampling Distribution:** The distribution of a statistic (e.g. $\bar{X}$) over repeated samples.
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Quick Exercise

- You sample 100 people's heights. Which is the parameter, and which is the statistic?
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8. Point Estimation & Confidence Intervals

Key Points

- **Point Estimation:**
 - $\bar{X}$ estimates μ .
 - $\hat{p}$ estimates the true proportion p .

- **Confidence Intervals:**
 - For large n with known σ :

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

- If σ unknown and n is small, use a **t-distribution** with $n - 1$ degrees of freedom.
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Quick Exercise

- A sample of size $n = 25$ from a normal population has $\bar{X} = 50$, $s = 4$. Construct a 95% CI for μ .
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Suggested 2-Hour Agenda

1. (5 min) Probability & Set Theory
2. (5 min) Operations with Events
3. (10 min) Conditional Probability & Independence
4. (10 min) Random Variables & Distributions

5. **(10 min)** Discrete vs. Continuous
 6. **(15 min)** Normal Distribution & Applications
 7. **(15 min)** Basic Statistical Inference
 8. **(20 min)** Estimation & Confidence Intervals
 9. **(30 min)** Extra Practice / Review
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Final Tips

- Focus on **key formulas**: conditional probability, expectation, variance, confidence intervals.
- Practice with **representative exercises** (binomial, normal, confidence intervals).
- Use **z-tables** / **t-tables** for normal or t-distributions.
- Maintain a simple **checklist**: Did you apply each formula correctly? Interpret results in context?

Good luck!